

Stage I: Mean self-distillation

Prompts

$$x \sim \mathcal{D}$$

Frozen BF16
teacher

y , top- k logits

$$L_{KD} = (T^2/N_{\text{valid}}) \sum m_{b,t} \text{KL_top-}k(p_{T} \parallel p_S)$$

Trainable FP4
student

Stage II: FP4 GRPO post-training

FP4 policy after
Mean self-distillation

Quantized rollouts
 $\{y_i\}_{i=1}^G$

Task rewards and
group advantages

Update policy
with GRPO

FP4 linear operator inside the policy

